

# MOBILE APPLICATION ENABLING CITIZEN REPORT STREET PROBLEMS

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**Abstract—** The lethargicness among the corporation in solving the issues and problems in the local residential areas have resulted in a drastic rise in the street problems like faulty street lights, electricity problem, bad conditioned roads, stagnant water and many in which most of the issues are not being addressed properly and some issues are not even being reported. Since the complaints registered by the residents regarding the problems faced in their street are not addressed properly, people feel unmotivated to make complaints and give voice for their peaceful living. The impact of the current state of issue in local residential areas might result in irritation and conflicts among local residents. The project aims at achieving an ideal state where residents do not feel irritated in reporting any issues faced and are assured of the rectification of their complaints in a timely manner.

**Keywords—** street management, pothole detection, complaint management,

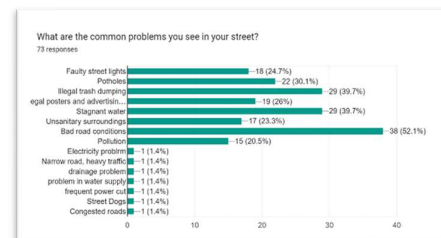
residents who are facing the day-to-day challenges and issues in their street/area. Most of the issues are not being addressed properly and some issues are not even being reported. Also, people feel difficulty in knowing the living conditions of the areas which are completely new to them when moving into a new area for residence.

## I. INTRODUCTION

Maintenance of the living habitat is a prime factor for a finer and healthy community living. This involves the cooperation of local residents in maintaining their surroundings and giving voice to any problems in their locality by reporting them in a timely manner. Also, involves the sincerity of respective governing departments in maintaining the welfare of the community and addressing the challenges and issues faced by the community people. Lack of awareness among the people about these local street issues may increase the lethargicness of the governing authorities responsible for street maintenance and peace. The project covered in this article seeks to provide local citizens with an easy way to lodge complaints. The paper makes an effort to enlighten and offer ideas to lessen the distance between citizens and government, bribery, and processing time.

## II. RESEARCH PROBLEM

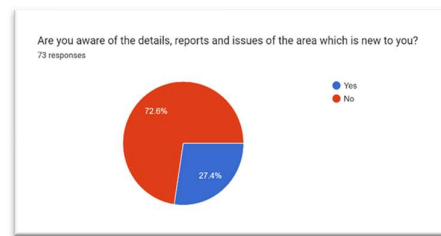
The context of the problem is Local residential areas with poor maintenance and increased rate of street issues like faulty street lights, electricity problem, bad conditioned roads, stagnant water and so on. Our target community is the local



(a) Common problems faced by local residents



(b) Addressal rate of the complaints registered by the user



(c) Awareness of living condition of areas among residents

Fig. 1. Real-time survey reports

The impact of the current state of issue in local residential areas might result in irritation and conflicts among local residents. Lack of awareness among the people about these local street issues may increase the lethargicness of the respective governing departments responsible for street maintenance and peace. Since the complaints registered by the residents regarding the problems faced in their street are not addressed properly, people feel unmotivated to make complaints and give voice for their peaceful living.

### III. RELATED WORKS

When examining the existing options, one of the obvious drawbacks is that none of the solutions permits feedback from the public; instead, complaints are simply closed by the government, leaving the public to believe that the problem has been fixed. Furthermore, while filing a complaint, citizens are unaware of how long it would take for a response. Additionally, since most solutions do not permit other users to vote complaints up or down, the government would not be under enough pressure.

Such solutions also provide the appearance that governments have enough resources to address all complaints made by citizens. It is also clear that there may be an abundance of data because even complaints without the necessary supporting documentation (pictures or videos) are sent to the system right away, which causes the arrival of numerous fake complaints. Also in the existing system, the filed complaints by the users were not in display to the public and people didn't know about the existing problems in a city which is completely new to them. Each fresh solution must consider these and lessen their effects. Some of the existing systems are discussed below.

**Maiaam Whistle mobile app:** Maiaam Whistle application was launched by Kamal Hasan on April 30. The application enables local residents of Tamilnadu and Puducherry to report any issues like corruption, pollution or other civic problems with proper evidential proofs. The verification is done by champions/party volunteers. Once the complaint is verified to be a Valid one. The file issue will be reported to the government and can be viewed by the public. However, the application filed complaints and is not much efficient in tracking the complaints to get it resolved and the application is only accessible to the registered members of Makkal Needhi Maiaam(MNM) and is not accessible to government officials. The last updation of the application is on January 7, 2019. And the application is only available for the older version of android.

**TN CM Citizen helpline app:** Tamilnadu CM helpline application is a Public Grievance CM Helpline Management System released on January 28, 2021. The application enables citizens to report local issues. The process of complaint registration is too complex and time consuming. The citizen can view only the problems reported by them and not the other problems. The review report says that the efficiency of the application in making the problem resolved and updating the user with the filed complaint is poor. The last updation of the application is on 15th November, 2022..

### IV. LITERATURE SURVEY

We have reviewed many research works to gain knowledge and insight into how our project is being implemented.

[1] **AI-Driven Complaint Management System**, Authors: Shreyas Shedge, Pratik Wadkar, Atreya Raorane, Bushra Shaikh, Year: 2021 - In this paper, the proposed model is an AI-driven logging portal where they try to improve communication between citizens and government. The designed website is coded in Sublime Text and has used HTML5, CSS3 and PHP for creating the website. For storing data, the system used XAMPP & MySQL and Python to perform operations on data. Matplotlib is used for showing the solved no. of cases and the unsolved no. of cases in graphical form. The project provides an overview of the AI-Driven complaint logging portal as a web service design and execution.

[2] **Pothole Detection using Accelerometer and Computer Vision with Automated Complaint Redressal**, Authors: Aditi Kandoi, Harsh Agarwal, Abhishek Revadekar, Bhalchandra N Chaudhari, Year: 2021- The system suggests an approach that employs a real-time pothole detection method to perform automatic pothole detection integrating accelerometer and gyroscope sensors inbuilt in smartphones as well as through computer vision. The proposed methodology involves gathering and analysis of this data, which the civic authorities in charge of maintaining the roads can then use this data. By user input, the technology aids the local government in keeping track of the number of potholes and enhancing road conditions.

[3] **Municipal Corporation Complaint Management System using web application**, Authors: Adharshini, Suvedha, Vanisri, Vaishnavi, Year: 2021- In this paper, the proposed system helps people to lodge a complaint with supporting documentation, such as photos of the things that are responsible for causing inconvenience and the locations will be monitored using GPS (Global Positioning System). With this guiding principle in mind, it describes four initiatives they undertook, 1. A cell phone campaign to address public complaints 2. Citizen-based oversight of government services 3. A system for managing complaints based on community input 4. A news service for rural communities over the phone.

[4] **Online Complaint Management System**, Authors: Lovely Singh Bhadouria, Nikunjay Kumar, Abdul Faisalc, Mrs. Suman Devid, Year: 2021 - With the Portal, one may file a complaint with higher government entities online. The majority of the Google APIs include Geocoding, Maps, Pin Code, and Firebase Cloud Informing services. The place selector offers a user interface (UI) that displays an intelligent map and a list of nearby places, including places compared to land locations and adjacent businesses. Customers may select a location, and the application can then get the points of interest associated with that placement. The programme makes use of a number of Google APIs to facilitate user interaction. The phone's GPS position is tracked and utilised as the complaint location by default when the user is prompted to choose a place.

[5] **All India Grievance Redressal App**, Author: Viral Patel, Daanyaal Kapadia, Deval Ghevariya, Shiburaj Pappu, Year: 2020- A conceptual framework that includes a citizen complaint management IT solution is built. A mobile app and a web app were used for the front-end implementation, allowing users to submit complaints quickly and simply along with images, videos, and GPS coordinates. Once a complaint has been resolved, residents may verify the resolution using photos, videos, and ratings. If they are not satisfied with the outcome, they can resubmit or escalate the complaint to a

different government official. Government institutions are currently supported by a web application that makes it simple to track all complaints and gives authorities the means to escalate issues between departments.

[6] Smart community maintenance ecosystem with self-reporting and self-diagnostic IoT sensors, Authors: Pravesh Kumar Tejan and Shilpi Kain, Year: 2021- This suggested system aims to address a number of problems, including street light control, rapid vehicle movement, tracking of occupied or hijacked parking spaces, overgrown green belt development, electrical problems, water management control, and IOT growth. Also, locals find it difficult to file complaints on various websites or applications. The goal of the paper is to offer an intelligent solution utilising IOT sensors. By the use of an automated backend for registering complaints, these sensors would automatically identify the problems and notify the relevant authorities. Also, this backend controls the status of the complaint register for future required reviews. The problem statement and its importance are covered at the outset of the study. The study also aims to describe how to create a sensor system that is self-healing and minimal maintenance. The paper also provides a deep understanding of IoT sensors.

## V. PROPOSED SYSTEM

The project aims at achieving an ideal state where residents do not feel irritated in reporting any issues faced and are assured of the rectification of their complaints in a timely manner. The project also focuses on making the public aware of the maintenance of each area by the respective corporations and current living conditions of all the residential areas. Thereby increasing the level of seriousness of addressing the issues among the respective departments and helping people, who are moving into a new location, choose an area that would be suitable for them to move in.

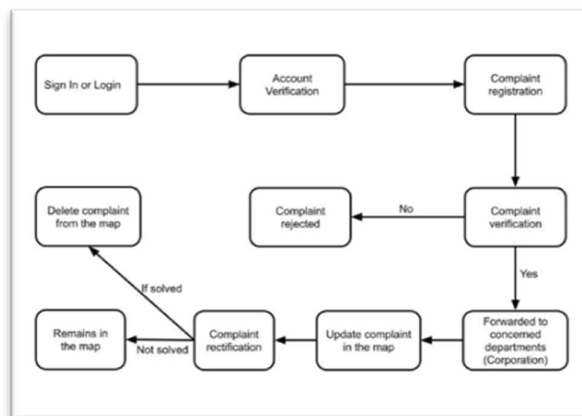


Fig. 2. Application Flow

The project is based on mobile application development. The application acts as a platform that enables the citizens to report the issues in their areas to the corporations and panchayats. The complaints posted by the citizens will be filed, notified, updated and viewed in the portal only after the process of verification of the complaint. The complaint will be sent to the respective corporation and panchayat authorities. The process of verification of Pothole complaints is auto-verified by pothole detection whereas the verification of other common complaints seeks the involvement of local NGOs. The public can view the current reports of any desired locations. Users may add photos and

comments to their complaint. The verified issues are displayed on the city's map. The public can view the current reports of any desired locations. The history of complaint records can be viewed by the authorities. Once the complaint is filed in a particular street, The street gets closer to the red zone. and the complaints are sent to the corporation and panchayat authorities.

## VI. APPLICATION FLOW

The user will first have to download the application proceeded by creating a user account. Users can login to the application using their respective username and password. The application has four logins, one login is for the users who are the residents, second login is for the local Non-Governmental Organizations, third login is for the corporation, who are able to observe the verified complaints and take necessary actions. and the final login is for the Admin.

The admin has the overall control over the application. and the NGOs login has the power to view, verify and approve the valid complaints to the respective authorities and corporations and delete the complaint once it has been addressed by the government properly. The user login has the power to file and view complaints. The user will first file a complaint with the following details, his name, address and other personal details then proceed with choosing the domain on which he has to file the complaint then will be adding descriptions and photos (optional for some issues) to the complaints.

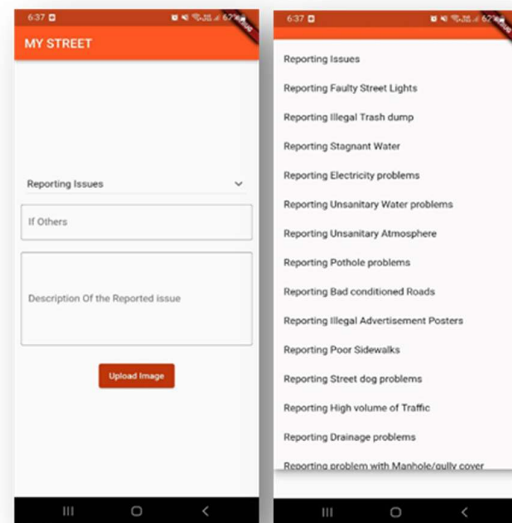


Fig. 3. Filing complaints

These complaints will now be notified to the local NGOs. The NGOs verify whether the filed complaint is a valid one or a fake one. The user account posting more than five fake complaints will be temporarily blocked by the admin and the user account posting more than seven fake complaints will be permanently blocked by the admin. The verified valid complaints will be forwarded to the corporations and authorities of relevant domains by the admin. These complaints will be notified to the authorities until the problem is solved. Once the complaints are filed, it will be in a public view and as the complaints get moving up stick, the particular area will be moving towards the red zoned area which will in turn increase the pressure of not addressing these issues properly which is now in a public view.

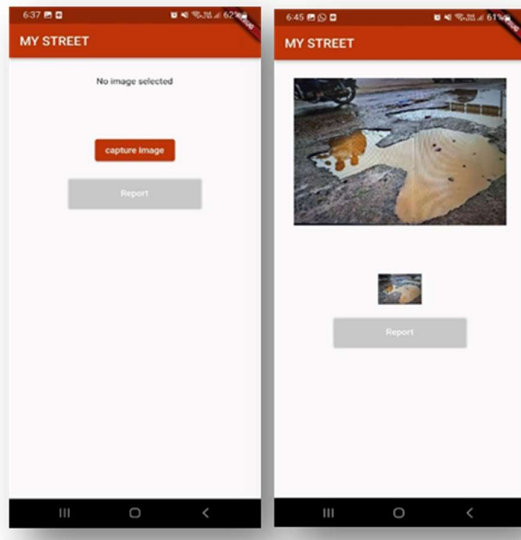


Fig. 4. Capturing live image and reporting

## VII. POTHOLE DETECTION

Pothole issues will be verified by Artificial intelligence and the verified complaints will be sent to local NGOs, corporations and panchayat authorities. whereas the verification of other common complaints seeks the involvement of local NGOs. To ease the process of verification of the complaints, we have developed a module for the automatic detection of the potholes.

The detection model is developed using deep learning object detection. Object detection process served under different variations in poses, occlusions, viewpoints, and lighting conditions of the input data. We have used the dataset size of 150 images of the pothole, 110 for training and 40 for testing. The database was trained with the average loss and learning rate. The precision and recall have been analysed using the evaluation process. The process of pothole detection includes data processing (data collection and data pre-processing), Splitting for testing and validation, Converting to COCO Format, implementing R-CNN, training and testing the model.

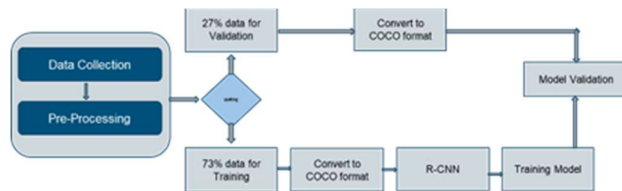


Fig. 5. Process flow – Pothole detection

### A. Image Classification

By using transfer learning to RCNN, a pre-trained CNN is used as a feature extractor, and the RCNN network is then tuned for a new object identification task. Usually, a big dataset like ImageNet, which has millions of photos and thousands of object categories, is used to train the pre-trained CNN. In order to perform a variety of computer vision tasks, including object detection, CNN has learned to extract generic visual features. The RCNN network can perform better on the new task with less training data and effort by utilising transfer learning. This is so that the network can concentrate on

learning the particular item because the pre-trained CNN has already learned to extract useful picture features, which can be employed in the current assignment. The following list of key steps represents the suggested training procedure:

1. Generating three separate sets out of the dataset (e.g., Training set, Validation set, and Testing set).
2. Establishing the initial values for the hyper-parameters (such as learning rate, number of epochs, weight decay, etc.).
3. Train the network using the starting values from step 2.
4. Utilising the validation set during the learning phase to evaluate the network performance.
5. 2000 epochs of step 3 and step 4 repetition.
6. Deciding on the model with the best training that has the lowest error rate on the validation set.
7. Using the testing set to determine the trained model's actual performance.

### B. Data Processing

#### • Training the Model:

In order to train the model, the operating system has been imported and we have set up the pre-trained model by downloading it from TensorFlow model zoo. The TensorFlow object detection repository had been cloned and TensorFlow object detection API was installed from the protocol buffer and was imported. The label map is then created for the labeling of the object. The TensorFlow record file format to train the object detection model is created. The model is then trained with the training steps of 2300.

The main goal of this dataset is to create a deep-learning model that can accurately identify potholes on roadways. The dataset comprises photographs of potholes that were captured with a smartphone camera and has a resolution of (500 x 500) pixels. The images were saved in JPG format. Fig. 6 shows some examples of the pothole classifications.



Fig. 6. Pothole Classifications

#### • Testing the Model:

To show the effectiveness of the suggested image segmentation and classification methodologies, numerous thorough tests are carried out. The fully trained model must have a very low amount of loss. In our case we have the loss of 0.131. The evaluation of the model has been done to measure the amount of precision and recall. These have to be high for the model to work effectively and efficiently. The



average precision is about 0.115 and the average recall is 0.202.

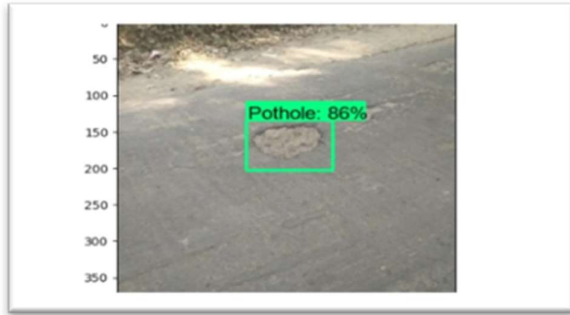


Fig. 7. Pothole Detection

## VIII. RESULT AND CONCLUSION

Our application provides users with a platform to report the problems being faced in their residential area and bring those problems to the public view. Thus, the system pressurizes the unsolved problem reported and these issues after verification will be notified to the respective departments and the municipalities. Regular updates on the action taken on these issues will be intimated to the respective reporter. Auto verification of potholes and trash dumping minimizes the workload of local NGOs. The system reduces the procedures and minimises the amount of time it takes to handle complaints. It also expands the channels available for filing complaints, progress reporting, and tracking complaint status.

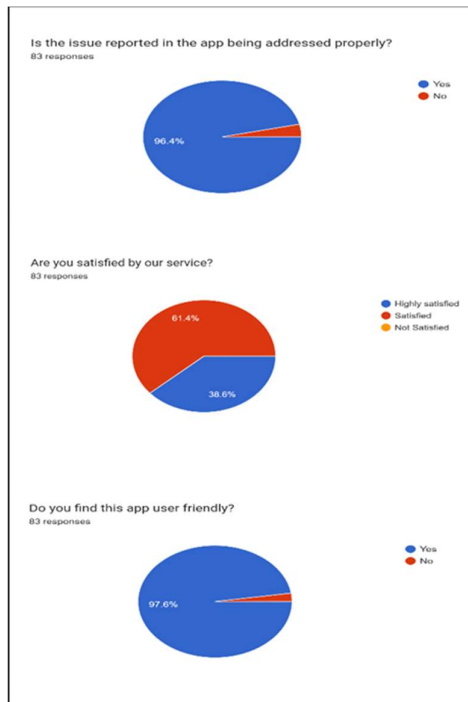


Fig. 8. Feedback Survey

## IX. FUTURE SCOPE

The system acts as the bridge between the public grievances and respective departments concerned with taking necessary measures. The Future vision of the project is to automate the process of verification of these complaints using Machine Learning instead of manual verification by the local

Non Governmental bodies, which may save time and effort without the involvement of third parties. Further research might be done on how complaints are routed to specific government agencies depending on the issues they raise, and it would be possible to incorporate sound logic that takes geolocation into account. Also, more studies could be done on how to lessen the impact of misuse, which can result in an excess of priority complaints for governmental organisations.

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